

Three's a Crowd: An Experimental Investigation Into Person Features

JANE MIDDLETON & SWARNENDU MOITRA

Queen Mary University of London

Abstract

Experimental investigations into the learnability of person and number syncretisms have found that not all syncretisms are equally learnable. Given traditional assumptions about feature systems, it is expected that learners will easily acquire syncretisms that arise when two items share a feature. We argue that it is easier to learn exponents for marked features than unmarked ones, and propose that markedness plays a role too. We present the results of an artificial language experiment in which participants were taught a pronominal quadripartition and one of six auxiliary tripartitions, which suggest that markedness does play a role in the learning of syncretisms.

Keywords: artificial language learning, markedness, number systems, person systems, pronouns, syncretism

1 Introduction

Across the world, no language has been discovered to have more than four denotations of person.¹ Moreover, the denotations of person are cross-linguistically stable: 1st person exclusive means ‘us but not you’ (i_o), 1st inclusive means ‘you and us’ (iu_o), 2nd means ‘you but not us’ (u_o), and the 3rd person means ‘others, neither you nor us’ (o_o). Logically, the four-person space can have 15 possible partitions: a monopartition, a quadripartition, seven bipartitions, and six tripartitions (Harbour 2016: 41). But of the possible 15, only five have been argued to exist: the monopartition, two bipartitions, a tripartition, and the quadripartition, Figure 1 (see Harbour 2016: §7.2.1 for an overview).

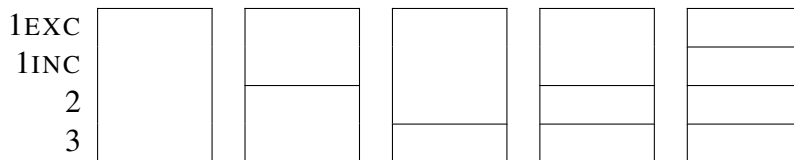


Figure 1: Attested partitions of the person space

A number of accounts have been developed to explain the limited variation in person systems (e.g. Harley & Ritter 2002; McGinnis 2005; Bobaljik 2008; Harbour 2016; Ackema & Neeleman 2018), each in some way encoding the intuitive idea that the 1st exclusive shares some property with the 1st inclusive, which in turn shares some property with 2nd. Yet this is despite Zwicky’s observation – made as far back as 1977 – that the 1st inclusive always coalesces with the 1st exclusive and never the 2nd when the person space has a bi- or tripartition.

The focus of the present paper is the derivations in Harbour 2016, which cut the person pie with the features [$\pm$ author] and [$\pm$ participant]. These operate on the person lattice, which consists of every possible combination of persons.² When [$\pm$ author] applies to the lattice, it divides the combinations

¹Five, if one includes the proximate and distal 3rd persons of obviative systems, but the difference between these is generally attributed to properties of proximity rather than person.

²See Harbour 2016 for a more technical exposition; my aim here is to present Harbour’s analysis with no more detail than is necessary to follow the rest of the arguments in this paper.

into those that include the speaker (i_o and iu_o) and those that do not (u_o and o_o). When $[\pm\text{participant}]$ applies, it divides the combinations into those that include both the speaker and addressee (i_o , iu_o and u_o), and those that do not (o_o). Consequently, Zwicky's observation falls out from Harbour's features; there is no way to group the 1st inclusive and 2nd persons together to the exclusion of 1st exclusive (Harbour 2016: 74).

Harbour's analysis was put to the test in Maldonado & Culbertson 2022. They ran an experiment in which participants were taught a fragment vocabulary from an artificial language. In each condition the person space was tripartite, with the 1st inclusive being syncretic with another person. Harbour derives the attested tripartition ($iu_o-u_o-o_o$) by applying $[\pm\text{author}]$ to the person lattice first, followed by $[\pm\text{participant}]$. The other tripartitions ($i_o-iu_o-o_o$ and $i_o-u_o-iu_o$) are underivable. If this analysis is correct, Maldonado & Culbertson anticipated that participants would find the fragment vocabulary in which the 1st exclusive and 1st inclusive were syncretic relatively easy to learn, but the ones in which the 1st inclusive was syncretic with the 2nd or 3rd person significantly harder to learn.

Contrary to expectations, however, the 1INC-2 syncretism was only marginally harder to learn than the 1EXC-1INC syncretism, and significantly easier to learn than the 1INC-3 syncretism. Maldonado & Culbertson concluded that this indicates that participants were sensitive to the fact that the 1st inclusive and 2nd persons share the feature $[+\text{participant}]$, and consequently found this syncretism easy to learn.

A caveat is missing here, however. In order for participants to ascertain that the 1st inclusive has coalesced with a person other than 1st exclusive, they must posit an underlying quadripartition, rather than a tripartition, because – if Harbour is right – the only tripartition available syncretises the 1st persons. The quadripartition arises from the person features applying to the person lattice in the reverse order ($[\pm\text{participant}]$, then $[\pm\text{author}]$). Harbour's derivations of the four persons are given in (1).

(1) *The person feature denotations* (Harbour 2016)

- a. 1EXC $[-\text{participant} +\text{author}]$
- b. 1INC $[+\text{participant} +\text{author}]$
- c. 2 $[+\text{participant} -\text{author}]$
- d. 3 $[-\text{participant} -\text{author}]$

A person tripartition can be derived from the quadripartition via underspecification. The 1EXC-1INC syncretism is derived through their shared feature $[+\text{author}]$, the 1INC-2 syncretism through $[+\text{participant}]$, and 1INC-3 through total underspecification of the person features. Since it's possible to derive any tripartition via the quadripartition and underspecification, it is unsurprising that Maldonado & Culbertson's participants had no trouble learning the 1INC-2 syncretism. But their struggle to learn the 1INC-3 syncretism remains puzzling; if underspecification is all that's needed, this syncretism should be just as learnable as the others.

We propose that markedness plays a role too. Specifically, we argue that exponents for marked features are easier to learn than exponents for unmarked ones, and that consequently, syncretisms derived therefrom will be easier to learn than those requiring the exponence of unmarked features. We point first to experimental evidence supporting our view from the number domain, §2, before presenting experimental data of our own in the person domain, §3. We conclude in §4.

2 Markedness In Number

Before we begin, we should briefly state our theoretical assumptions. The analysis in this paper is presented within the framework of Distributed Morphology (Halle & Marantz 1993; Halle 1997). In this framework, the terminal nodes of trees are spelt out and sent to PF, where they receive their

phonological form. The phonological forms of the terminal nodes are determined by rules of exponence, which are found on the language’s Exponent List. The rules of exponence are governed by the Subset Principle (Kiparsky 1973; Halle & Marantz 1993; Halle 1997), which states that for a given bundle of features, the exponent that realises the maximal subset of that bundle of features must be used. We’ve seen the person features already; the number features are $[\pm\text{atomic}]$ and $[\pm\text{minimal}]$ (Harbour 2014).

Harbour 2014 derives number distinctions by applying number features to the number lattice.³ $[\pm\text{atomic}]$ applies to the lattice first; $[+\text{atomic}]$ selects a single item, and $[-\text{atomic}]$ selects a non-singular item. Then $[\pm\text{minimal}]$ applies; $[+\text{minimal}]$ selects the smallest set of items compatible with the set already selected, and $[-\text{minimal}]$ selects a set of items that is at least one more than the set already selected.

(2) *The number feature denotations* (Harbour 2014)

- a. Singular $[+\text{atomic} +\text{minimal}]$
- b. Dual $[-\text{atomic} +\text{minimal}]$
- c. Plural $[-\text{atomic} -\text{minimal}]$

Harbour writes the semantic content of the features to ensure that the root must first combine with $[\pm\text{atomic}]$, and then this combination can combine with $[\pm\text{minimal}]$, (3).

(3) *The semantic content of Harbour’s features*

- a. $[\pm\text{atomic}] = \lambda x (\neg)\text{atom}(x)$
- b. $[\pm\text{minimal}] = \lambda P \lambda x (\neg)\neg\exists y(P(y) \wedge y \sqsubset x)$
presupposition: $P(x)$

Now consider the data in Table 1 (Smith *et al* 2019: Appendix B2), which shows the number paradigms for twelve persons in one Central Solomons language (Lavukaleve) and five Australian languages (Djamindjung, Nyigina, Ngandi, Warrwa and Wunambal).

³See Harbour 2014 for a more technical exposition; my aim here is to present Harbour’s analysis with no more detail than is necessary to follow the rest of the arguments in this paper.

DL	yirri- nji	gurri- nji	e- l	me- l
PL	yirri	gurri	e	me
SG	ɲayug	nami	ngai	ngai
	1EXC	2	1EXC	1INC
	Djamindjung		Lavukaleve	

DL	yarrdju- mirri	gurrnga- mirri	yirrnga- mirri	ɲorrkorr- ni
PL	yarrdju	gurrnga	yirrnga	ɲorrkorr
SG	ɲayu	djuwa	ginja	njaka
	1INC	2	3	1INC
	Nyigina		Ngandi	

DL	yaara- wili	kurra- wili	yirra- wili	njarra- miya
PL	yaara	kurra	yirra	njarra
SG	ngayu	juwa	kinya	ɲaya
	1EXC	2	3	1EXC
	Warrwa		Wunambal	

Table 1: Data suggesting that the marked value of $[\pm\text{minimal}]$ is positive

Each of the paradigms exhibits the same morphological structure. There is a unique morpheme for the singular person, a shared morpheme for the dual and plural persons, and an additional morpheme in the dual. This can be captured with an exponent that spells out the relevant person features and $[+\text{atomic}]$ for the singular person, another that spells out the relevant person features and $[-\text{atomic}]$ for the morpheme shared between dual and plural, and an exponent for the feature that distinguishes dual from plural, $[+\text{minimal}]$. We’ve illustrated this for Djamindjung 1EXC pronouns in (4).

- (4) *Exponents for Djamindjung 1EXC pronouns*
 $[-\text{participant} +\text{author} +\text{atomic}] \Leftrightarrow \text{ɲayug}$
 $[-\text{participant} +\text{author} -\text{atomic}] \Leftrightarrow \text{yirri}$
 $[+\text{minimal}] \Leftrightarrow \text{-nji}$

Next, consider the data in Table 2 (Smith *et al* 2019: Appendix B2), which shows the number paradigms for five persons in two Austroasiatic languages (Semelai and Mlabri), an Australian language (Mangarayi), and a Bantu language (Lega-Shabunda).

PL	hɛ- ʔen	bah jum	rnur- la	b -íswé	bá -bo
DL	hɛ	bah	rnurr	íswé	bo
SG	ʔɛɲ	mɛh	ɲianggi	nne	gwě
	1INC	2	2	1	2
	Semelai	Mlabri	Mangarayi	Lega-Shabunda	

Table 2: Data suggesting that the marked value of $[\pm\text{minimal}]$ is negative

Each of these paradigms shows an identical morphological structure to those in Table 1, but with the dual and plural persons switched: there is a unique morpheme for the singular person, a shared morpheme for the dual and plural persons, and an additional morpheme in the plural. This data can be captured with an exponent that spells out the relevant person features and [+atomic] for the singular person, another that spells out the relevant person features and [−atomic] for the morpheme shared between dual and plural, and an exponent for the feature that distinguishes plural from dual, [−minimal]. We’ve illustrated this for Semelai 1INC pronouns in (5).

(5) *Exponents for Semelai 1INC pronouns*

- [+participant +author +atomic] ⇔ ?əŋ
 [+participant +author −atomic] ⇔ he
 [−minimal] ⇔ -?en

In all seventeen paradigms we find a morphological division between the singular persons and the dual and plural persons. This is as expected, as we find that, cross-linguistically, the primary pair of natural classes in the number domain are singular and non-singular. This is hinted at in Universal 34: *No language has a trial number unless it has a dual. No language has a dual unless it has a plural* (Greenberg 1963). Universal 34 makes the observation that there is a number hierarchy, such that if a language distinguishes number, trial is rarer than dual, and dual is rarer than plural. The number hierarchy is actually a series of binary oppositions. The first divides the number domain into the singular and the non-singular; the second divides the non-singular domain into the minimally non-singular (the dual) and the non-minimally non-singular (the plural); and the third divides the minimally non-singular domain into the augmented minimally non-singular (the trial) and the non-augmented minimally non-singular (the dual). Since the first opposition is between the singular and the non-singular, the primary number feature is [±atomic]. It’s a binary feature because its values denote complementary natural classes.

The marked value of [±atomic] is universally accepted to be negative: *There is no language in which the plural does not have some non-zero allomorphs, whereas there are languages in which the singular is expressed only by zero. The dual and the trial are almost never expressed only by zero* (Universal 35, Greenberg 1963). Markedness is the expression of a cross-linguistic inequality between two items in a complementary pair. For example, the unmarked item is more widely distributed throughout the world’s languages; the presence of the marked item in a given domain implies the presence of the unmarked one, but not *vice versa*; and when both items are active in a language, the unmarked item appears first in child language acquisition (Harris 1994: 92). The property of markedness that concerns us here is exponence: marked values of features are more likely to be expounded than unmarked ones. Since all languages that distinguish singular and plural expound the plural in some way while the singular may remain unexpounded, plural is marked with respect to singular, making the marked value of [±atomic] negative. This analysis is consolidated by the fact that plural follows singular in the number hierarchy.

The second opposition in the number domain divides the non-singular domain into the minimally non-singular (the dual) and the non-minimally non-singular (the plural). This division is captured by the binary feature [±minimal]; [+minimal], when combined with [−atomic], denotes the natural class of dual items, while [−minimal], when combined with [−atomic], denotes the natural class of pluralities greater than two. However, there’s no equivalent to Universal 35 to indicate what the marked value of [±minimal] is. That dual follows plural in the number hierarchy suggests that it’s positive, but the morphological evidence is ambiguous on this point: the data in Table 1 (where [+minimal] is expounded) supports this view, while the data in Table 2 (where [−minimal] is expounded) suggests that it’s negative.

To illustrate the order in which the number features combine, Smith *et al* 2019 – who first brought the data in Tables 1 and 2 to general attention – represent them in a tree structure with the number lattice (#) at the bottom, $[\pm\text{atomic}]$ just above it, and $[\pm\text{minimal}]$ just above that, Figure 2.

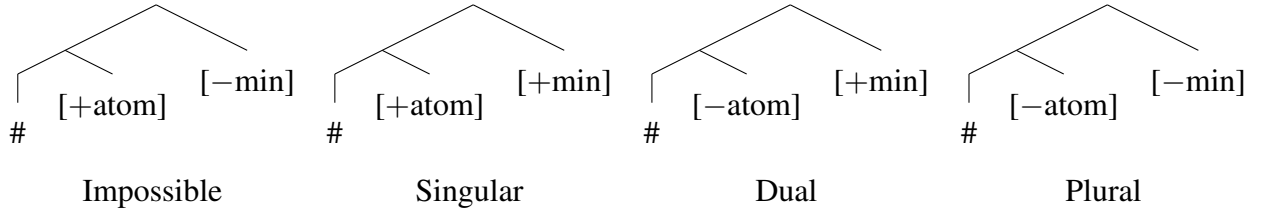


Figure 2: Smith *et al*'s 2019 trees

Smith *et al* discuss number syncretisms with abstract AXY notation: AAA, AAB, ABA, ABB and ABC. These labels arise from assumptions about the marked values of $[\pm\text{atomic}]$ and $[\pm\text{minimal}]$, and the exponents that spell out the three (possible) trees. We will briefly digress to elucidate these assumptions, as we make use of the AXY notation throughout the rest of the paper.

2.1 A Digression on AXY Labels

When Bobaljik 2012 brought *ABA suppletion to general notice, he was discussing adjectival triplets. The comparative and superlative features are privative; they are either present or absent. Consequently the tree structures differ in size, Figure 3.

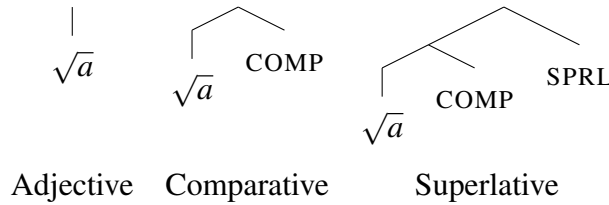


Figure 3: Adjective-Comparative-Superlative trees, with privative features

When the features are privative and the tree structures cumulative, the AXY label corresponds to the growing tree: AAA refers to a root that is exponed by the same exponent when it's in the adjectival tree, the comparative tree, and the superlative tree. ABB refers to a root that has one form when it appears in the adjectival tree, but a suppletive form when it appears in the comparative and superlative trees. ABC refers to a root that has one form when it appears in the adjectival tree, a suppletive form in the comparative tree, and another suppletive form in the superlative tree.

As mentioned earlier, the values of binary features denote complementary natural classes. Since the present paper is discussing binary number features in equally sized trees, the AXY label corresponds to the markedness of the features that spell out the trees. The A number is that denoted by the unmarked value of the first feature to apply to the number lattice and the value of second feature that is unmarked in the context of the first's unmarked value; the B number is that denoted by the marked value of the first feature to apply to the number lattice and the value of the second that is unmarked in the context of the first's marked value; and the C number is that denoted by the marked value of the first feature to apply to the number lattice and the value of the second feature that is marked in the context of the first's marked value. There are four logically possible ways to derive an ABC pattern for number.

In the first derivation, the marked value of $[\pm\text{atomic}]$ is negative, and the marked value of $[\pm\text{minimal}]$ is positive, (6). ABC corresponds to SG-PL-DL, because singular is denoted by the

two unmarked values of $[\pm\text{atomic}]$ and $[\pm\text{minimal}]$, plural by the marked value of $[\pm\text{atomic}]$ and unmarked value of $[\pm\text{minimal}]$, and dual by the marked values of both $[\pm\text{atomic}]$ and $[\pm\text{minimal}]$.

(6) *The exponents of marked features in the first derivation*

$[\# +\text{minimal} -\text{atomic}] \Leftrightarrow \text{C}$	<table><tr><td>C</td></tr></table>	C	DL
C			
$[\# -\text{atomic}] \Leftrightarrow \text{B}$	<table><tr><td>B</td></tr></table>	B	PL
B			
$[\#] \Leftrightarrow \text{A}$	<table><tr><td>A</td></tr></table>	A	SG
A			

In the second derivation, the marked value of $[\pm\text{atomic}]$ is negative, and the marked value of $[\pm\text{minimal}]$ is negative, (7). ABC corresponds to SG-DL-PL, because singular is denoted by the two unmarked values of $[\pm\text{atomic}]$ and $[\pm\text{minimal}]$, dual by the marked value of $[\pm\text{atomic}]$ and the unmarked value of $[\pm\text{minimal}]$, and plural by both the marked values of $[\pm\text{atomic}]$ and $[\pm\text{minimal}]$.

(7) *The exponents of marked features in the second derivation*

$[\# \text{ --minimal --atomic}] \Leftrightarrow \text{C}$	<table><tr><td>C</td></tr></table>	C	PL
C			
$[\# \text{ --atomic}] \Leftrightarrow \text{B}$	<table><tr><td>B</td></tr></table>	B	DL
B			
$[\#] \Leftrightarrow \text{A}$	<table><tr><td>A</td></tr></table>	A	SG
A			

These derivations are compatible with Smith *et al*'s 2019 trees in Figure 2 because the first feature that applies to the number lattice is $[\pm\text{atomic}]$, and in these derivations its marked value is negative.

The remaining two derivations are compatible with trees where the first feature that applies to the number lattice is $[\pm\text{minimal}]$, followed by $[\pm\text{atomic}]$ (this combination order is incompatible with Harbour's semantics), Figure 4.

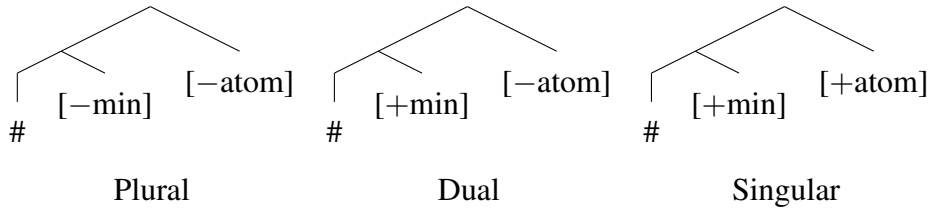


Figure 4: Smith *et al*'s 2019 trees with the number features inverted

In the third derivation, the marked value of $[\pm\text{minimal}]$ is positive, and the marked value of $[\pm\text{atomic}]$ is positive, (8). ABC corresponds to PL-DL-SG, because plural is denoted by the unmarked values of $[\pm\text{atomic}]$ and $[\pm\text{minimal}]$, dual by the marked value of $[\pm\text{minimal}]$ and the unmarked value of $[\pm\text{atomic}]$, and singular by both the marked values of $[\pm\text{atomic}]$ and $[\pm\text{minimal}]$.

(8) *The exponents of marked features in the third derivation*

$[\# +\text{atomic} +\text{minimal}] \Leftrightarrow \text{C}$	<table><tr><td>C</td></tr></table>	C	SG
C			
$[\# +\text{minimal}] \Leftrightarrow \text{B}$	<table><tr><td>B</td></tr></table>	B	DL
B			
$[\#] \Leftrightarrow \text{A}$	<table><tr><td>A</td></tr></table>	A	PL
A			

In the fourth derivation, the marked value of $[\pm\text{minimal}]$ is positive, and the marked value of $[\pm\text{atomic}]$ is negative, (9). ABC corresponds to PL-SG-DL, because plural is denoted by the unmarked values of

$[\pm\text{atomic}]$ and $[\pm\text{minimal}]$, singular by the marked value of $[\pm\text{minimal}]$ and the unmarked value of $[\pm\text{atomic}]$, and dual by both the marked values of $[\pm\text{atomic}]$ and $[\pm\text{minimal}]$.

(9) *The exponents of marked features in the fourth derivation*

$[\# \text{ --atomic } +\text{minimal}] \Leftrightarrow \text{C}$	<table><tr><td>C</td></tr></table>	C	DL
C			
$[\# +\text{minimal}] \Leftrightarrow \text{B}$	<table><tr><td>B</td></tr></table>	B	SG
B			
$[\#] \Leftrightarrow \text{A}$	<table><tr><td>A</td></tr></table>	A	PL
A			

Since these derivations are incompatible with the semantics of Harbour’s features and fail to capture the insight that the primary number feature is $[\pm\text{atomic}]$, we will not consider them any more.

The important point here is that the AXY label applies to the markedness of the features that spell out the trees, and not to the numerical value of singular, dual and plural. Which order singular, dual and plural come in with respect to the AXY label is determined by two things: the order in which the number features apply to the number lattice, and which values of those features are marked.

2.2 Experimental predictions for number

The experiments discussed in this paper teach participants a fragment vocabulary of an artificial language which features a syncretism between two numbers or two persons. In both cases two binary features combine to denote four categories (one combination of which is semantically impossible in the number domain). In the domain of number, Table 3, a syncretism between two of the three categories can be captured in up to two ways.

	$+\text{atomic}$	$-\text{atomic}$
$+\text{minimal}$	SG	DL
$-\text{minimal}$	-	PL

Table 3: The number feature grid

A syncretism between singular and dual can be captured with an exponent for their shared feature, $[\text{+minimal}]$, and an elsewhere exponent for the plural, (10).

(10) *Deriving the SG-DL syncretism*

$$[\# +\text{minimal}] \Leftrightarrow \text{SG} / \text{DL}$$

$$[\#] \Leftrightarrow \text{PL}$$

(11) *Deriving the SG-DL syncretism*

$$[\# -\text{minimal} -\text{atomic}] \Leftrightarrow \text{PL}$$

$$[\#] \Leftrightarrow \text{SG} / \text{DL}$$

Alternatively, it could be captured with an exponent for the plural features, $[-\text{minimal} -\text{atomic}]$, and an elsewhere exponent for the syncretic singular and dual, (11).

A syncretism between dual and plural can be captured with an exponent for their shared feature, $[-\text{atomic}]$, and an elsewhere exponent for the singular, (12).

(12) *Deriving the DL-PL syncretism*

$$\begin{aligned} [\# -\text{atomic}] &\Leftrightarrow \text{DL} / \text{PL} \\ [\#] &\Leftrightarrow \text{SG} \end{aligned}$$

(13) *Deriving the DL-PL syncretism*

$$\begin{aligned} [\# +\text{atomic}] &\Leftrightarrow \text{SG} \\ [\#] &\Leftrightarrow \text{DL} / \text{PL} \end{aligned}$$

Alternatively, it could be captured with an exponent for the singular features, [+minimal +atomic], and an elsewhere exponent for the syncretic dual and plural, (13).

Finally we have the SG-PL syncretism. There is only one pair of exponents that will capture this, (14), as singular and plural have no feature in common.

(14) *Deriving the SG-PL syncretism*

$$\begin{aligned} [\# +\text{minimal} -\text{atomic}] &\Leftrightarrow \text{DL} \\ [\#] &\Leftrightarrow \text{SG} / \text{PL} \end{aligned}$$

The null hypothesis is therefore that all three syncretisms will be equally easy to learn.

Recall that marked values of features are more likely to be expounded than unmarked ones. We hypothesise that learners easily infer exponents for marked values of features, but find it harder to infer exponents for unmarked ones. Thus we predict that syncretisms that arise via the underspecification of marked feature values will be easier to learn than syncretisms derived another way. If one of the syncretism pairs is significantly harder to learn than the other two, then our assumptions about markedness and exponence are supported; if the syncretisms are equally easy to learn, then we have evidence that markedness doesn't play a role in exponence. Since we already know that the first number feature to apply to the lattice is [$\pm$ atomic], and its marked value is negative, we only have two possibilities to consider.

If the marked value of [$\pm$ minimal] is positive, we predict that the SG-PL and DL-PL syncretisms will be easy to learn, Table 4.

	ABC	AAB	ABB
DL	$[\# +\text{min} -\text{atom}] \Leftrightarrow \text{C}$	$[\# +\text{min} -\text{atom}] \Leftrightarrow \text{B}$	$[\# -\text{atom}] \Leftrightarrow \text{B}$
PL	$[\# -\text{atom}] \Leftrightarrow \text{B}$	$[\#] \Leftrightarrow \text{A}$	$[\#] \Leftrightarrow \text{A}$
SG	$[\#] \Leftrightarrow \text{A}$		
		SG-PL	DL-PL

Table 4: Exponents for AAB and ABB if the marked value of [$\pm$ minimal] is positive

Note that if the marked value of [$\pm$ minimal] is positive, the AXY order is SG-PL-DL.

If the marked value of [$\pm$ minimal] is negative, we predict that the SG-DL and DL-PL syncretisms will be easy to learn, Table 5.

	ABC	AAB	ABB
PL	$[\# -\text{min} -\text{atom}] \Leftrightarrow \text{C}$	$[\# -\text{min} -\text{atom}] \Leftrightarrow \text{B}$	$[\# -\text{atom}] \Leftrightarrow \text{B}$
DL	$[\# -\text{atom}] \Leftrightarrow \text{B}$	$[\#] \Leftrightarrow \text{A}$	$[\#] \Leftrightarrow \text{A}$
SG	$[\#] \Leftrightarrow \text{A}$		
		SG-DL	DL-PL

Table 5: Exponents for AAB and ABB if the marked value of [$\pm$ minimal] is negative

Note that if the marked value of $[\pm\text{minimal}]$ is negative, the AXY order is SG-DL-PL.

Since the DL-PL syncretism is predicted to be easy to learn regardless of the marked value of $[\pm\text{minimal}]$, it is the SG-DL and SG-PL syncretisms that are of particular interest.

2.3 The experiment presented in Lee 2021

Lee 2021 describes an artificial language experiment in which participants were taught one of three fragment vocabularies, each of which included one of the three number syncretisms. Participants were introduced to two friendly aliens and instructed to try to learn their language. The experiment was conducted online, with the aliens and their equipment presented visually, and the artificial language presented aurally.

In the first training phase, participants were taught the names of the six fruits (*feek*, ‘apple’; *deet*, ‘banana’; *lun*, ‘cherry’; *dite*, ‘lemon’; *seeg*, ‘pear’; *pem*, ‘carrot’) and three nominal number suffixes used in the experiment. These suffixes remained constant across all conditions (*-cha*, ‘singular’; *-po*, ‘dual’; *-fi*, ‘plural’). In the first testing phase, participants completed a forced-choice referent selection task, in which they heard the audio for a fruit with its number suffix, and had to select the correct quantity from three given images.

In the second training phase, participants were taught the verbs *pim*, ‘beam up’, and *bice*, ‘throw away’, and two verbal number suffixes, *-te* and *-ku*. In one condition *-te* denoted both singular and dual agreement, in another it denoted both singular and plural agreement, and in the third, *-ku* denoted dual and plural agreement. In the second testing phase, participants completed another forced-choice referent selection task. The alien on the ground told the alien in the spaceship how to make a smoothie; the participants’ job was to listen to the instructions and select the correct number of fruits to be beamed up or thrown away by the alien in the spaceship. Their task was hindered by the fact that some of the audio was obscured.

The alien’s instructions took the form of verb-noun imperatives (e.g. *Pim-te feek-cha!*, ‘Beam up an apple!’), with the disambiguating number-inflected nouns rendered inaudible by the obscuring noise. For each instruction, participants were presented with the relevant fruit in two quantities: singular or dual, singular or plural, or dual or plural. Each pair of options was given for both verbal number suffixes, excepting the syncretic pair, which only appeared with the correct verbal suffix. Participants thus faced four test items per fruit (for a total of four fruits). 149 participants successfully completed Lee’s experiment, with 52 assigned to the DL-PL condition, 49 to the SG-DL condition, and 48 to the SG-PL condition.

2.4 Lee’s 2021 results

Since the nominal number suffixes were constant across all conditions, participants were tested on identical data in the first testing phase. Participants’ accuracy in the DL-PL condition was significantly above chance, at approximately 63% (chance = 33.3%; $p < 0.001^{***}$), and neither the participants in the SG-PL condition nor the SG-DL condition were significantly different from the DL-PL baseline ($p = 0.336$ and 0.147 , respectively).

The experimental results varied in the second testing phase. Participants’ accuracy in the DL-PL condition was well above chance, at 68.6% (chance = 50%; standard error = 0.034, $p < 0.001^{***}$), while participants’ accuracy in the SG-PL condition was lower, but still above chance, at 60.7% (standard error = 0.030, $p = 0.0567$). However, the participants’ accuracy in the SG-DL condition was significantly lower than the baseline, performing close to chance at 51.8% (standard error = 0.033, $p < 0.001^{***}$).

2.5 Discussion of the results in Lee 2021

Lee's results confirm that participants are most successful at learning the SG-PL and DL-PL syncretisms. This indicates that the marked value of $[\pm\text{minimal}]$ is positive, corroborating Universal 34: *No language had a trial number unless it has a dual. No language has a dual unless it has a plural* (Greenberg 1963). Dual follows plural in the number hierarchy, and $[+\text{minimal}]$ is the feature that distinguishes dual in the non-singular domain.

Recall, however, that the DL-PL syncretism could be accounted for by two pairs of exponents. The first pair had an exponent for the marked $[-\text{atomic}]$, (12), and the second had an exponent for the unmarked $[+\text{atomic}]$, (13). One could argue that we cannot say for certain that markedness plays a part in explaining language acquisition, since Lee's participants could have inferred either pair of exponents when learning this syncretism.

(12) *Deriving the DL-PL syncretism*

$[\# -\text{atomic}] \Leftrightarrow \text{DL} / \text{PL}$

$[\#] \Leftrightarrow \text{SG}$

(13) *Deriving the DL-PL syncretism*

$[\# +\text{atomic}] \Leftrightarrow \text{SG}$

$[\#] \Leftrightarrow \text{DL} / \text{PL}$

To weigh up the strength of this argument, let's look more closely at the pairs of exponents that could account for the SG-DL syncretism, which Lee's participants struggled to learn.

The first pair of exponents consists of an exponent for $[+\text{minimal}]$, the feature shared by singular and dual, and an elsewhere exponent for plural, (10). The syncretism is derived via underspecification of $[-\text{atomic}]$. The SG-PL syncretism is also derived via underspecification of $[-\text{atomic}]$, but as participants had no trouble learning it, this cannot explain why the SG-DL syncretism was so much harder to learn. This must therefore be due to some property of the expounded feature $[+\text{minimal}]$.

$[+\text{minimal}]$ is marked in the presence of $[-\text{atomic}]$ (the feature bundle for dual), and participants had no trouble learning this exponent in the SG-PL condition. By contrast, $[+\text{minimal}]$ is trivially true in the presence of $[+\text{atomic}]$, making it unmarked in the singular. Thus, the exponents in (10) involve the exponence of an unmarked feature.

(10) *Deriving the SG-DL syncretism*

$[\# +\text{minimal}] \Leftrightarrow \text{SG} / \text{DL}$

$[\#] \Leftrightarrow \text{PL}$

(11) *Deriving the SG-DL syncretism*

$[\# -\text{minimal} -\text{atomic}] \Leftrightarrow \text{PL}$

$[\#] \Leftrightarrow \text{SG} / \text{DL}$

The second pair of exponents consists of an exponent for plural and an elsewhere exponent for singular and dual, (11). The syncretism is derived via underspecification of singular's unmarked features, and dual's marked ones. As was the case with the exponents in (10), an unmarked feature is expounded: plural's $[-\text{minimal}]$. It appears, then, that participants in the SG-DL condition found it difficult to infer the exponents in (10) and (11) because each pair expounded an unmarked value of a feature.

Greenberg's Universal 35 states that *There is no language in which the plural does not have some non-zero allomorphs, whereas there are languages in which the singular is expressed only by zero. The dual and the trial are almost never expressed only by zero*. Universal 35 implies that the derivation of the DL-PL syncretism in (13) is never inferred by speakers. Because it is possible to distinguish the singular from the dual and plural by the unmarked feature $[+\text{atomic}]$ alone, we could conclude that unmarked features are never expounded.

However, we know that unmarked features can be expounded (e.g. the exponents for $[+\text{minimal}]$ in Table 2, (5)), so the derivation of the DL-PL syncretism in (13) cannot be completely impossible. We must reach a more restrained conclusion then, namely that given the choice of two pairs of exponents that derive the data, speakers do not entertain the derivation of syncretisms involving the exponence of unmarked features, when a derivation only involving the exponence of marked features is available.

Lee’s results therefore support the hypothesis that learners easily infer exponents for marked values of features, but find it harder to infer exponents for unmarked ones. This furthers our understanding of markedness, showing that natural classes cannot on their own account for the acquisition of morphemes; markedness plays an important role too.

3 Markedness in Person

Now let’s look at person. To recap, Harbour 2016 derives the person distinctions by applying person features to the person latticee, π , which denotes all person combinations. The quadripartition is derived when the first person feature that is applied to the lattice is $[\pm\text{participant}]$, followed by $[\pm\text{author}]$. The four persons have the structures in Figure 5.

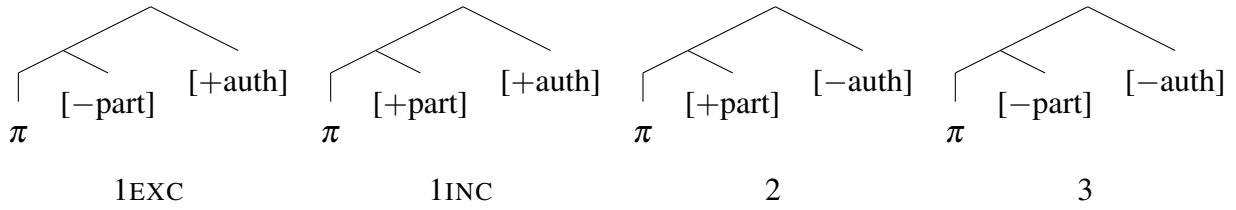


Figure 5: The tree structures of person

3.1 Experimental predictions for person

Person is more complicated than number because no combination of features is semantically impossible, Table 6. This means that there is only one way to derive each of the six logically possible syncretisms.

	+author	−author
+participant	1INC	2
−participant	1EXC	3

Table 6: The person feature grid

A syncretism between 1EXC and 1INC can be captured with an exponent for their shared feature, $[+\text{author}]$, and two exponents fully specified for 2nd and 3rd person, (15).

(15) *Deriving the 1EXC-1INC syncretism*

$$[\pi -\text{auth} -\text{part}] \Leftrightarrow 3$$

$$[\pi -\text{auth} +\text{part}] \Leftrightarrow 2$$

$$[\pi +\text{auth}] \Leftrightarrow 1\text{EXC} / 1\text{INC}$$

(16) *Deriving the 1EXC-2 syncretism*

$$[\pi -\text{auth} -\text{part}] \Leftrightarrow 3$$

$$[\pi +\text{auth} +\text{part}] \Leftrightarrow 1\text{INC}$$

$$[\pi] \Leftrightarrow 1\text{EXC} / 2$$

The 1EXC and 2nd persons have no feature in common, so a syncretism between them can be captured with an elsewhere exponent and two exponents fully specified for 1INC and 3rd person, (16).

A syncretism between the 1EXC and 3rd persons can be captured with an exponent for their shared feature, $[-\text{participant}]$, and two exponents fully specified for 1INC and 2nd person, (17).

(17) *Deriving the 1EXC-3 syncretism*

$$[\pi - \text{auth} + \text{part}] \Leftrightarrow 2$$

$$[\pi + \text{auth} + \text{part}] \Leftrightarrow 1\text{INC}$$

$$[\pi - \text{part}] \Leftrightarrow 1\text{EXC} / 3$$

(18) *Deriving the 1INC-2 syncretism*

$$[\pi - \text{auth} - \text{part}] \Leftrightarrow 3$$

$$[\pi + \text{auth} - \text{part}] \Leftrightarrow 1\text{EXC}$$

$$[\pi + \text{part}] \Leftrightarrow 1\text{INC} / 2$$

A syncretism between the 1INC and 2nd persons can be captured with an exponent for their shared feature, [+participant], and two exponents fully specified for 1EXC and 3rd person, (18).

The 1INC and 3rd persons have no feature in common, so a syncretism between them can be captured with an elsewhere exponent and two exponents fully specified for 1EXC and 2nd person, (19).

(19) *Deriving the 1INC-3 syncretism*

$$[\pi - \text{auth} + \text{part}] \Leftrightarrow 2$$

$$[\pi + \text{auth} - \text{part}] \Leftrightarrow 1\text{EXC}$$

$$[\pi] \Leftrightarrow 1\text{INC} / 3$$

(20) *Deriving the 2-3 syncretism*

$$[\pi + \text{auth} + \text{part}] \Leftrightarrow 1\text{INC}$$

$$[\pi + \text{auth} - \text{part}] \Leftrightarrow 1\text{EXC}$$

$$[\pi - \text{auth}] \Leftrightarrow 2 / 3$$

A syncretism between the 2nd and 3rd persons can be captured with an exponent for their shared feature, [−author], and two exponents fully specified for 1EXC and 1INC, (20).

As with number, the null hypothesis is that all six syncretisms will be equally easy to learn. We retain our hypothesis that learners easily infer exponents for marked values of features, but find it harder to infer exponents for unmarked ones, again predicting that syncretisms that arise via the underspecification of marked feature values will be easier to learn than syncretisms derived another way. In the domain of person, the first feature to apply to the person lattice is [±participant], but it is not known which value is marked.

The second feature to apply to the person lattice is [±author]. As it's the dependent feature, its marked value is contextually determined. It's not known if only one value is marked in the context of both values of [±participant], or if one value is marked in the context of [+participant], while the other is marked in the context of [−participant]. However, if the marked value of [±author] does not vary with the value of [±participant], two syncretisms will be derivable with exponents for marked features, whereas if it does, only one of the two can be derived. We address this in §3.1.5. We thus have four different markedness combinations to explore.

(21) *The four possible combinations of person markedness in the quadripartition*

- a. The marked value of [±participant] is positive, and the marked value of [±author] is positive in that context
- b. The marked value of [±participant] is positive, and the marked value of [±author] is negative in that context
- c. The marked value of [±participant] is negative, and the marked value of [±author] is positive in that context
- d. The marked value of [±participant] is negative, and the marked value of [±author] is negative in that context

The AXY order of the persons is calculated thusly. The A person is that denoted by the unmarked value of [±participant] and the value of [±author] that is unmarked in that context. There are two B persons: the one that is denoted by the marked value of [±participant] and the value of [±author] that is unmarked in that context, and the one that is denoted by the unmarked value of [±participant] and the value of [±author] that is marked in that context. The C person is that denoted by the marked value of [±participant] and the value of [±author] that is marked in that context.

	$[^m\text{auth}] \not\Leftarrow$	$[^m\text{part}] \not\Leftarrow$	
C	$[\pi \text{ } ^m\text{auth } ^m\text{part}] \Leftrightarrow \text{C}$	$[\pi \text{ } ^m\text{auth } ^m\text{part}] \Leftrightarrow \text{C}$	C
B	$[\pi \text{ } ^m\text{part}] \Leftrightarrow \text{B}$	$[\pi \text{ } ^m\text{auth}] \Leftrightarrow \text{B}$	B
A	$[\pi] \Leftrightarrow \text{A}$	$[\pi] \Leftrightarrow \text{A}$	A
	A-A	A-A	

Table 7: Syncretisms expected for the marked values $[^m\text{part}]$ and $[^m\text{auth}]$

The first syncretism arises via underspecification of the marked value of $[\pm\text{author}]$, while the second arises via underspecification of the marked value of $[\pm\text{participant}]$. Beneath the exponents are the syncretisms they correspond to.

3.1.1 Predictions for Combination 1

If the marked values of $[\pm\text{participant}]$ and $[\pm\text{author}]$ are positive, then the two syncretisms that should be easy to learn are 1EXC-3 and 2-3.

	$[+\text{auth}] \not\Leftarrow$	$[+\text{part}] \not\Leftarrow$	
1INC	$[\pi \text{ } +\text{auth } +\text{part}] \Leftrightarrow \text{C}$	$[\pi \text{ } +\text{auth } +\text{part}] \Leftrightarrow \text{C}$	1INC
2	$[\pi \text{ } +\text{part}] \Leftrightarrow \text{B}$	$[\pi \text{ } +\text{auth}] \Leftrightarrow \text{B}$	1EXC
1EXC	$[\pi] \Leftrightarrow \text{A}$	$[\pi] \Leftrightarrow \text{A}$	2
3	1EXC-3	2-3	3

Table 8: Syncretisms expected if $[+\text{participant}]$ and $[+\text{author}]$ are marked

3.1.2 Predictions for Combination 2

If the marked value of $[\pm\text{participant}]$ is positive, and the marked value of $[\pm\text{author}]$ is negative, then the two syncretisms that should be easy to learn are 1EXC-3 and 1EXC-1INC.

	$[-\text{auth}] \not\Leftarrow$	$[+\text{part}] \not\Leftarrow$	
2	$[\pi \text{ } -\text{auth } +\text{part}] \Leftrightarrow \text{C}$	$[\pi \text{ } -\text{auth } +\text{part}] \Leftrightarrow \text{C}$	2
1INC	$[\pi \text{ } +\text{part}] \Leftrightarrow \text{B}$	$[\pi \text{ } -\text{auth}] \Leftrightarrow \text{B}$	3
3	$[\pi] \Leftrightarrow \text{A}$	$[\pi] \Leftrightarrow \text{A}$	1INC
1EXC	1EXC-3	1EXC-1INC	1EXC

Table 9: Syncretisms expected if $[+\text{participant}]$ and $[-\text{author}]$ are marked

3.1.3 Predictions for Combination 3

If the marked value of $[\pm\text{participant}]$ is negative, and the marked value of $[\pm\text{author}]$ is positive, then the two syncretisms that should be easy to learn are 1INC-2 and 2-3.

	[+auth] ✗	[-part] ✗	
1EXC	$[\pi +auth -part] \Leftrightarrow C$	$[\pi +auth -part] \Leftrightarrow C$	1EXC
3	$[\pi -part] \Leftrightarrow B$	$[\pi +auth] \Leftrightarrow B$	1INC
1INC	$[\pi] \Leftrightarrow A$	$[\pi] \Leftrightarrow A$	3
2	1INC-2	2-3	2

Table 10: Syncretisms expected if [-participant] and [+author] are marked

3.1.4 Predictions for Combination 4

If the marked values of [$\pm$ participant] and [$\pm$ author] are negative, then the two syncretisms that should be easy to learn are 1INC-2 and 1EXC-1INC.

	[-auth] ✗	[-part] ✗	
3	$[\pi -auth -part] \Leftrightarrow C$	$[\pi -auth -part] \Leftrightarrow C$	3
1EXC	$[\pi -part] \Leftrightarrow B$	$[\pi -auth] \Leftrightarrow B$	2
2	$[\pi] \Leftrightarrow A$	$[\pi] \Leftrightarrow A$	1EXC
1INC	1INC-2	1EXC-1INC	1INC

Table 11: Syncretisms expected if [-participant] and [-author] are marked

3.1.5 Predictions when the marked value of [$\pm$ author] varies

We mentioned earlier that only one syncretism can be derived with exponents for marked features if the marked value of [$\pm$ author] varies with the value of [$\pm$ participant]. We demonstrate this here by showing that if the marked value of [$\pm$ participant] is negative and the marked value of [$\pm$ author] is negative in that context, but positive in the context of [+participant] (*cf.* Combination 4), the predicted easy-to-learn syncretism is 1INC-2. With these three marked features, C expone the 3rd person and A expone 2nd. The two B persons are thus 1EXC and 1INC; 1EXC is denoted by the marked [-participant] and unmarked [+author] features, while 1INC is denoted by the unmarked [+participant] and marked [+author] features.

	[+auth] ✗	[-part] ✗	
3	$[\pi -auth -part] \Leftrightarrow C$	$[\pi -auth -part] \Leftrightarrow C$	3
1EXC	$[\pi -part] \Leftrightarrow B$	$[\pi +auth] \Leftrightarrow B$	1EXC
1INC	$[\pi] \Leftrightarrow A$	$[\pi] \Leftrightarrow A$	1INC
2	1INC-2	1EXC-1INC	2

Table 12: Syncretisms expected if different values of [$\pm$ author] are marked in different contexts

If there is no exponent for the marked [+author] feature, the syncretism derived is 1INC-2.

If there is no exponent for the marked [-participant] feature, and there is one for the marked [+author] one, the 1EXC-1INC is derived. In this instance it will be an ABBC pattern because the B exponent ([+author]) will surface for both 1EXC and 1INC, leaving A to expone the 2nd person. There is a problem here, however: [+author] is unmarked in the 1EXC. We argued earlier that participants find it hard to infer exponents for features that are unmarked, even when in other contexts they are marked (see (10) in §2.5), so we predict this syncretism to be difficult to learn here too.

We leave it to the reader to verify that only one syncretism is predicted to be easy to learn whenever the marked value of $[\pm\text{author}]$ varies with the value of $[\pm\text{participant}]$.

3.1.6 Summary of the Predictions

The predicted syncretisms reveal a pattern. If the marked value of $[\pm\text{participant}]$ is positive, the syncretism that should be easy to learn is 1EXC-3, while if it is negative, the syncretism is 1INC-2. If the marked value of $[\pm\text{author}]$ is positive, the syncretism that should be easy to learn is 2-3, while if it is negative, it's 1EXC-1INC. If participants learn only the 1EXC-3 or the 1INC-2 and no other, we take this to indicate that the marked value of $[\pm\text{author}]$ varies with the value of $[\pm\text{participant}]$.

We also predict that neither the 1EXC-2 nor the 1INC-3 syncretisms should be easy to learn. This is because participants must infer that all four features are expounded (see (16) and (19)), contra our assumption that it's easier to infer exponents for marked features than unmarked ones.

3.2 Person Experiment

The experiment we ran was a replication and expansion of Experiment 2 in Maldonado & Culbertson 2022. It differed from theirs in one way: following Lee 2021, the syncretism was hosted by an auxiliary verb rather than the pronoun. Participants were told that Harry, Ron and Hermione had found a magical hat which caused the wearer to speak a mysterious language, and they were tasked with helping the trio learn the language. They were taught one of six fragment vocabularies, each of which included a person syncretism. The experiment consisted of two training-and-testing phases.

3.2.1 Materials

The materials for the first training-and-testing phase consisted of 18 pairs of pictures. The picture on the left showed Harry, Ron and Hermione, one of whom was asking a question in English, and one of whom was wearing the magical hat, Figure 6. The picture on the right showed the wearer of the magical hat answering the question in the mysterious language, with the denotation of the word highlighted in colour, while the rest of the image was black and white. In this phase, participants learnt the three singular pronouns, 1st, 2nd, and 3rd. The nonce pronouns were *hoat*, *dain* and *melk*; their denotations varied between conditions. The questions always concerned events in the past, and the answers were always one word long.

The materials for the second training phase consisted of 24 pairs of pictures. Again, the picture on the left showed Harry, Ron and Hermione, one of whom was asking a question, and one of whom was wearing the magical hat, and the picture on the right likewise denoted the wearer of the magical hat answering in the mysterious language, with the meaning highlighted in colour and the rest of the picture in black and white, Figure 8. In this phase, participants learnt the four plural pronouns, 1EXC, 1INC, 2nd and 3rd. The nonce pronouns were *fich*, *rusp*, *woux* and *zeth*; their denotations also varied between conditions. The second phase differed from the first in that the questions concerned speculation about the future, and each pronoun was accompanied by one of three auxiliaries: *wan*, *wim* or *wung*, 'will'. These auxiliaries played host to the syncretism taught in each of the six conditions.

The materials for the second testing phase included a further 72 pictures, three for each answer. Each depicted the correct answer, but the speech bubble featured one of the three auxiliaries and an ink blot obscuring the pronoun, Figure 9. The pictures used in the experiment were created on StoryboardThat (<https://www.storyboardthat.com>).

3.2.2 Procedure

The experiment was run online using Gorilla software (<https://gorilla.sc/>) and the Prolific crowd-sourcing platform (<https://www.prolific.co/>). Participants were told their task was to help Harry, Ron and Hermione learn the mysterious language, and that they would be tested on their ability to learn it following training.

Phase 1: Training

The first training phase consisted of 18 trials; each trial had three screens. On the first screen, participants were shown a pair of pictures. The picture on the left showed one of Harry, Ron or Hermione asking a question, and the one on the right showed the person wearing the magical hat answering it, Figure 6.



Figure 6: Training in Phase 1

Underneath each pair was a text box; participants were instructed to type the mysterious word uttered by the person wearing the hat. This had to be an exact match (no spelling errors) for the participant to proceed to the next screen.

On the second screen, participants were shown another picture in which one of Harry, Ron or Hermione asks a question, underneath which are two possible answers; the answer to the question

on the first screen, and the correct answer to the new question. Participants were asked to click on the correct answer to the new question, whereupon they saw a third screen showing them the correct answer. The 18 trials were presented in three continuous blocks of six. Within each block of six, the trials were randomised. Each block of six contained each pronoun twice on the first screen, with the alternatives once each on the second screen.

Phase 1: Testing

The testing phase also consisted of 18 trials, one for each pronoun-alternative pair. In each trial, participants were shown a picture in which Harry, Ron or Hermione asks a question, Figure 7.



Figure 7: Testing in Phase 1

Underneath were two pictures depicting the answer. In one, the answer in the speech bubble was correct, and in the other it was incorrect. Participants had to click on the picture with the correct answer to the question.

The 18 trials were presented in three continuous blocks of six. Within each block of six, the trials were randomised. Each block of six contained each pronoun twice; the incorrect alternatives were given once per pronoun. The correct answers were distributed equally on the left and right.

Phase 2: Training

Participants who scored 70% or more in the first test progressed to the second training phase. The second training phase was identical to the first, Figure 8, but with 24 trials, instead of 18. The 24 trials were presented in six continuous blocks of four. Within each block of four, the trials were randomised. Each block contained every pronoun on the first screen once; the alternative pronouns were answers to the question on the second screen twice per correct pronoun over the 24 trials.

Phase 2: Testing

The testing phase consisted of 48 trials, two for each pronoun-alternative pair. In each trial, participants were shown a picture in which Harry, Ron or Hermione asks a question, Figure 9. Underneath were two pictures depicting the answer, but with an ink blot covering the pronouns in the speech bubbles. The auxiliary was correct in one picture, but incorrect in the other. Participants had to click on the picture with the correct answer to the question.

The 48 trials were in twelve continuous blocks of four. Within each block of four, the trials were randomised. Each block contained a question-answer pair for each pronoun; for every two blocks, the alternative auxiliaries were seen once per correct non-syncretic auxiliary, and twice per syncretic auxiliary. The correct answers were distributed equally on the left and right.

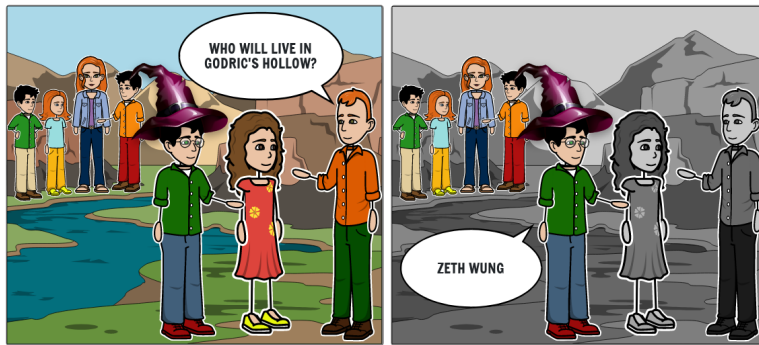


Figure 8: Training in Phase 2



Figure 9: Testing in Phase 2

3.2.3 Participants

813 participants were recruited via Prolific. Their data was included for analysis if they achieved a score of 70% or more in the first testing phase. This pass mark was set as it ensured the participants who progressed had learnt the meaning of the three singular pronouns. The figure is calculated thus: since there were six trials per pronoun, and three pronouns, participants needed to answer 13 (6 + 6 + 1) trials correctly for us to be sure they'd learnt them. 12/18 is 67%, and 13/18 is 72%, so the pass mark was set at 70%. This reduced the pool of participants to 377, of which 55 were assigned to the 1EXC-1INC condition, 60 to the 1EXC-2 condition, 73 to the 1EXC-3 condition, 58 to the 1INC-2 condition, 62 to the 1INC-3 condition, and 69 to the 2-3 condition.

Participants were between 18 and 30 years of age and were self-reported monolingual English speakers. Participants were randomly assigned to a condition by Gorilla. They were paid £1.20 for their participation, and awarded a bonus payment of £1.25 for every test they passed with a mark of 70% or more. Prolific has a minimal payment rate of £6 per hour (10p per minute), based on the average time all participants take to complete a study. As participants who scored 70% or more in the first test progressed to the second training and testing phase, we paid them a bonus of £1.25 to compensate them for the extra time they took and bring their payment up to a rate of £9 per hour (15p per minute). We offered another bonus of £1.25 for anyone who passed the second test with 70% or more, to encourage them to continue trying their hardest to learn the mysterious language and compensate them accordingly. The average time taken to complete the experiment was 12 minutes.

3.3 Results

Mean accuracy rates on test trials during the critical phase (Phase 2) are given in Figure 10⁴ and Table 13. The dashed line in Figures 10 and 11 indicate chance value (two options, forced choice).

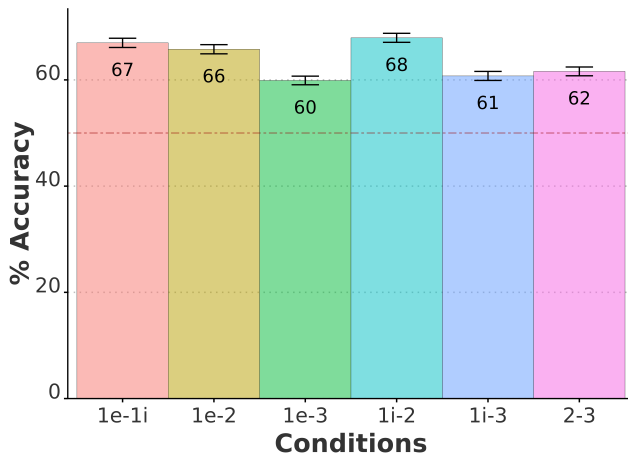


Figure 10: Mean accuracy by Condition plot

Conditions	Mean Accuracy (%)	Standard Error
1EXC-1INC	67.0	0.869
1EXC-2	65.8	0.863
1EXC-3	59.9	0.812
1INC-2	67.9	0.849
1INC-3	60.7	0.849
2-3	61.6	0.827

Table 13: Mean accuracy and standard error

All analysis was done using the R (v4.4.0) programming language (R Core Team 2022). To analyze the effect of conditions on accuracy rates, we used generalized linear mixed-effects models with a logit link function (Jaeger, 2008) using the lme4 package (Bates *et al* 2015). The models included random intercepts for both items and participants.⁵

⁴In the graphs, 1e = 1EXC and 1i = 1INC.

⁵Model equation: $Accuracy \sim Condition + (1 | Participant) + (1 | Item)$.

We first compared the 1EXC-1INC with the other Conditions, treating the 1EXC-1INC as the baseline as this Condition is English-like. The model intercept was significant ($\beta = 0.821$, $p < 0.001$), indicating that accuracy in this Condition was above chance. Compared to the 1EXC-1INC Condition, accuracy was significantly lower in the 1EXC-3 Condition ($\beta = -0.38$, $p = 0.003$), 1INC-3 Condition ($\beta = -0.33$, $p = 0.013$) and 2-3 Condition ($\beta = -0.30$, $p = 0.021$), Figure 11.

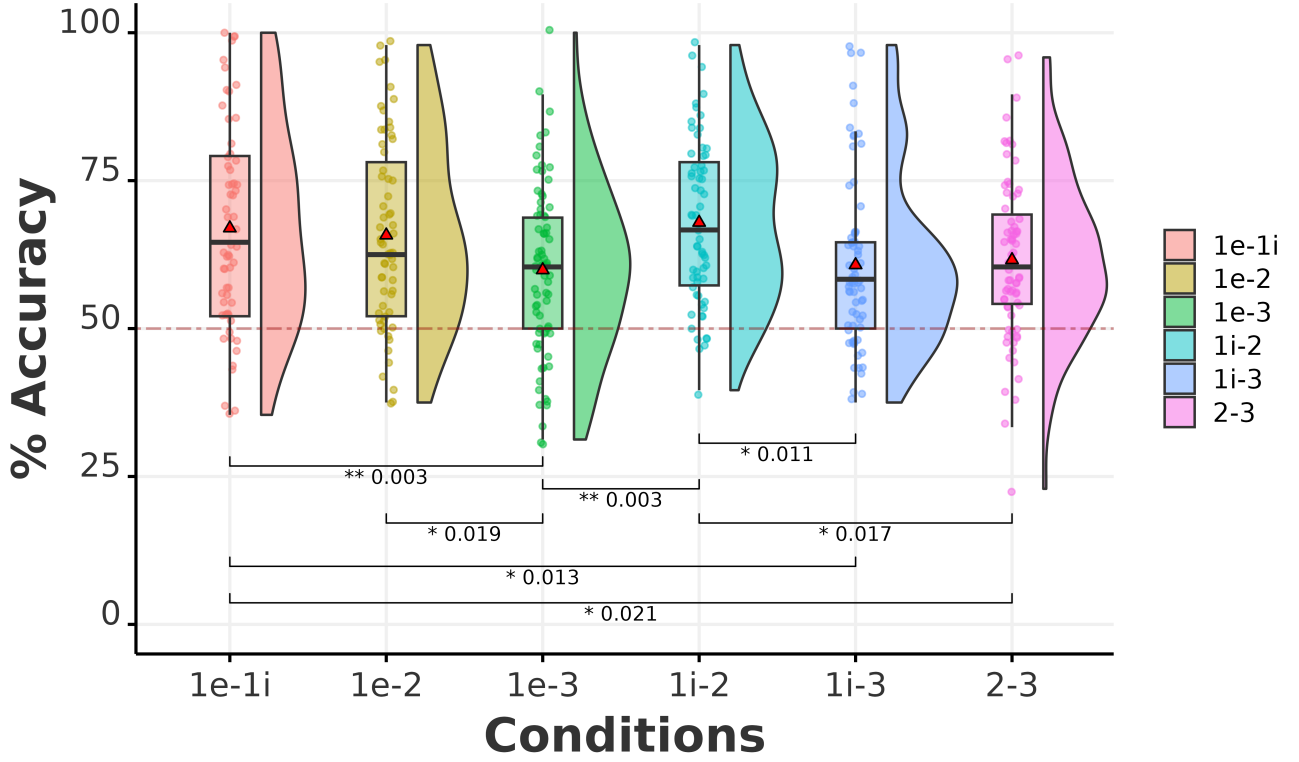


Figure 11: Mean Accuracy by Condition Plot

Following Maldonado & Culbertson 2022, successive models were built where the contrasts were treatment coded with each condition taken as a baseline, Table 14.

1EXC-2	0.557				
1EXC-3	**0.003	*0.019			
1INC-2	0.967	0.525	**0.003		
1INC-3	*0.013	0.057	0.681	*0.011	
2-3	*0.021	0.085	0.535	*0.017	0.841
	1EXC-1INC	1EXC-2	1EXC-3	1INC-2	1INC-3

Table 14: P-values when each Condition is treated as a baseline

The model intercepts were significant for all six Conditions, indicating that accuracy in each Condition was above chance (1EXC-1INC, ; 1EXC-2, $\beta = 0.741$, $p < 0.001$; 1EXC-3, $\beta = 0.442$, $p < 0.001$; 1INC-2, $\beta = 0.827$, $p < 0.001$; 1INC-3, $\beta = 0.493$, $p < 0.001$; 2-3, $\beta = 0.518$, $p < 0.001$).

Compared to the 1INC-2 Condition, accuracy was significantly lower in the 1EXC-3 Condition ($\beta = -0.384$, $p = 0.003$), 1INC-3 Condition ($\beta = -0.334$, $p = 0.011$) and 2-3 Condition ($\beta = -0.309$, $p = 0.017$), Figure 11 and Table 14.

Compared to the 1EXC-2 Condition, accuracy was significantly lower in the 1EXC-3 Condition ($\beta = -0.300$, $p = 0.019$), and marginally lower in the 1INC-3 Condition ($\beta = -0.249$, $p = 0.057$) and 2-3 Condition ($\beta = -0.223$, $p = 0.085$), Figure 11 and Table 14.

3.4 Discussion

These results confirm that participants were most successful at learning a new language that featured a 1EXC-1INC, 1EXC-2, or 1INC-2 syncretism. We predicted that if the marked value of $[\pm\text{participant}]$ was positive, participants would find the 1EXC-3 syncretism easy to learn (§3.1.1 and §3.1.2), while if it was negative, participants would find the 1INC-2 syncretism easy to learn (§3.1.3 and §3.1.4). Since participants were significantly better at learning the 1INC-2 syncretism than the 1EXC-3 one, our results support the claim that the marked value of $[\pm\text{participant}]$ is negative.

Our second prediction was that if the marked value of $[\pm\text{author}]$ was positive, participants would find the 2-3 syncretism easy to learn (§3.1.1 and §3.1.3), while if it was negative, participants would find the 1EXC-1INC syncretism easy to learn (§3.1.2 and §3.1.4). Since participants were significantly better at learning the 1EXC-1INC syncretism than the 2-3 one, our results support the claim that $[-\text{author}]$ is marked in the context of $[-\text{participant}]$.

Thirdly, we anticipated that if the marked value of $[\pm\text{author}]$ varies with the value of $[\pm\text{participant}]$, only one syncretism (1EXC-3 or 1INC-2) would be easily learnt. Since this wasn't the case, we might conclude that the marked value of $[\pm\text{author}]$ is constant, irrespective of the value of $[\pm\text{participant}]$. However, our fourth prediction went entirely awry. We expected both the 1EXC-2 and 1INC-3 syncretisms to be difficult to learn, as both require inferring that all four features are exponed, marked or not. One possible explanation is that we were wrong in predicting that only one syncretism would be easily learnt if the marked value of $[\pm\text{author}]$ varies with the value of $[\pm\text{participant}]$.

Since our participants' success at learning the 1EXC-1INC and 1INC-2 syncretisms suggests that $[-\text{participant}]$ is marked, and $[-\text{author}]$ is marked in that context, let's see how we'd derive the three easily learnt syncretisms if $[\text{author}]$ is marked in the context of $[\text{participant}]$. Compare the relevant tree structures for person with those for number, Figure 12; the marked values of features are in black.

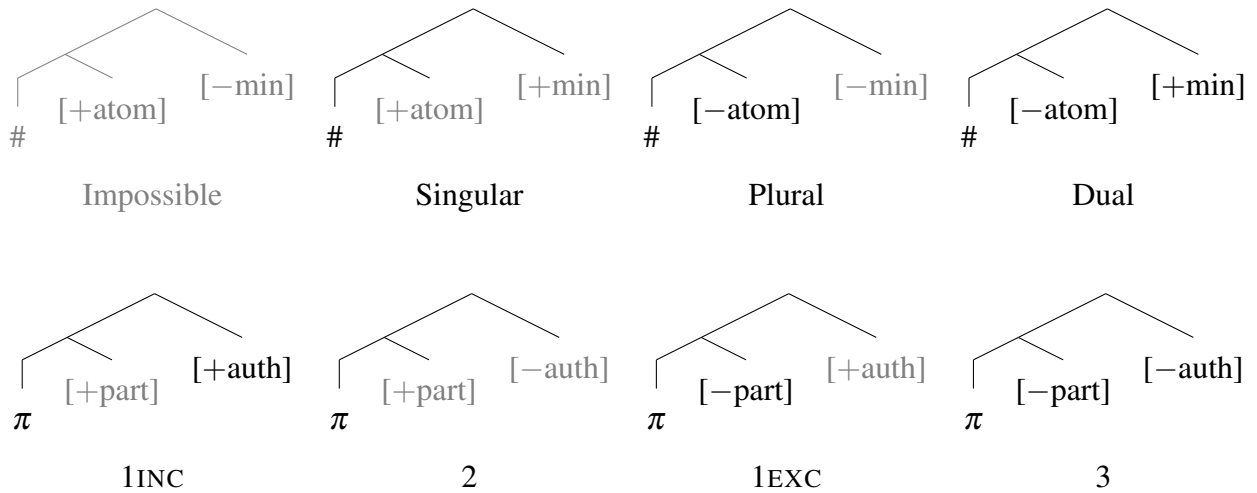


Figure 12: The structures of number and person, with marked features in black

In §3.1.5, we ruled out the derivation of the 1EXC-1INC syncretism which expones $[\text{author}]$ because this feature is marked when belonging to the 1st inclusive, and unmarked in the 1st exclusive. Lee's participants struggled to learn the SG-DL syncretism, which could be derived with an exponent for

[+minimal], which is marked in the dual and unmarked in the singular; thus we concluded such an exponent would be hard for participants to infer.

But the 1EXC-1INC can also be derived with an exponent for 2nd person's [−author] feature. This is unmarked here, but marked in the 3rd person, where it is exponed. Could it be that when learners infer an exponent for a marked feature, they find it easier to infer other exponents for the same feature, even when it is unmarked?

We need a different explanation to account for the learnability of the 1EXC-2 syncretism. Here we need to say that if learners can infer an exponent for a marked dependent feature, this makes it easier for them to infer one for the unmarked feature it is dependent on. This would account for our participants inferring an exponent for 1INC's features, even though [+participant] is unmarked.

These are speculative thoughts, though, so we hesitate to endorse this account. It's possible that the experimental design simply was not fine-grained enough to tease apart the learnability differences between the conditions, and a subtler design is required. It's also possible that there is some additional factor at play, hitherto unanticipated by anyone, which accounts for the three-by-three division in the learning of person syncretisms. Since we do not know what this might be, we will leave this puzzle for future research.

4 Conclusion

Tripartitions of the person space can be derived a number of different ways, once by applying [$\pm$ author] and then [$\pm$ participant] to the person lattice, and six ways with the reverse order of application. By deriving the tripartition from the quadripartition, we can account for the puzzling results in Maldonado & Culbertson 2022. Their participants were able to learn the 1INC-2 syncretism with the same degree of success as the 1EXC-1INC syncretism because they posited an underlying quadripartition and derived the two tripartitions via underspecification by inferring exponents only for marked features. They struggled to learn the 1INC-3 syncretism because doing so required them to infer exponents for unmarked features, which is a harder task.

Our key assumption was that learners find it easier to infer exponents for marked features than for unmarked ones. This was supported by experimental evidence in the number domain; participants in Lee 2021 learnt artificial languages with singular-plural and dual-plural syncretisms more successfully than those with a singular-dual syncretism, despite the fact that singular shares a feature with dual, but not with plural. These results are unexpected on an account in which shared features underlie learnable syncretisms, but expected on our account.

We applied this reasoning to the person domain, teaching participants an artificial language with a pronominal quadripartition and one of six auxiliary tripartitions. We found that participants were most successful in learning the 1EXC-1INC, 1EXC-2 and 1INC-2 syncretisms. Their ability to learn the latter indicates that the marked value of [$\pm$ participant] is negative, and their ability to learn the former indicates that the marked value of [$\pm$ author] is negative in that context. Our participants' success in learning the 1EXC-2 syncretism was unexpected. We tentatively suggested that it could be that when learners infer an exponent for one of a marked-unmarked pair, they find it easier to infer an exponent for the other one as well, but we left this open for further research.

More broadly, we believe our results demonstrate that experimental research on artificial languages presents a fruitful method for developing our understanding of feature systems and the way they operate.

References

- Ackema, P. & Neeleman, A. 2018. *Features of person: From the inventory of persons to their morphological realization* (Vol. 78). MIT Press.
- Bates, D. Mächler, M. Bolker, B. & Walker, S. 2015. *Fitting linear mixed-effects models using lme4*, J. Stat. Softw, 67(1), 1-48.
- Bobaljik, J. 2008. Missing persons: A case study in morphological universals. *The Linguistic Review* 25:203–230.
- Bobaljik, J. 2012. *Universals in comparative morphology: Suppletion, superlatives, and the structure of words*. MIT Press.
- Greenberg, J. 1963. Some universals of grammar with particular reference to the order of meaningful elements. *Universals of language*, 2, 73-113.
- Halle, M. & Marantz, A. 1993. Distributed morphology and the pieces of inflection.
- Halle, M. 1997. Distributed Morphology: Impoverishment and fission. *MIT Working Papers in Linguistics 30: Papers at the Interface*. Edited by B. Bruening, Y. Kang & M. McGinnis, 425–449. Cambridge, Mass. MITWPL.
- Harbour, D. 2014. Paucity, abundance, and the theory of number. *Language*, 90(1), 185-229.
- Harbour, D. 2016. *Impossible persons* (Vol. 74). MIT Press.
- Harley, H. & E. Ritter. 2002. Person and number in pronouns: A feature-geometric analysis. *Language* 78:482–526.
- Harris, J. 1994. *English sound structure*. Blackwell.
- Jaeger, T. F. 2008. Categorical data analysis: Away from ANOVAs (transformation or not) and towards logit mixed models. *Journal of memory and language*, 59(4), 434-446.
- Kiparsky, P. 1973. "Elsewhere" in phonology.
- Lee, N. 2021. *Learning (im)possible number syncretisms: investigating innate featural representations*. Manuscript under review.
- Maldonado, M., & Culbertson, J. 2022. Person of interest: Experimental investigations into the learnability of person systems. *Linguistic Inquiry*, 53(2), 295-336.
- McGinnis, M. 2005. On markedness asymmetries in person and number. *Language* 81:699–718.
- Smith, P. W. Moskal, B. Xu, T. Kang, J. & Bobaljik, J. 2019. Case and number suppletion in pronouns. *Natural Language & Linguistic Theory*, 37, 1029-1101.
- Zwicky, A. 1977. Hierarchies of person. In *Papers from the Thirteenth Regional Meeting, Chicago Linguistic Society*, ed. by Woodford A. Beach, Samuel E. Fox, & Shulamith Philosoph, 714–733. Chicago: University of Chicago, Chicago Linguistic Society.